

CSC241 - Object Oriented Programming (Java)

Pre-Lab Environment Setup Guide

For CS and AI Sections

READ THIS BEFORE COMING TO THE LAB. Complete the setup on your own laptop before the 3-hour lab. The purpose is to arrive ready to write, compile, run and manage Java code - not to spend the lab downloading software.

Internet note: You will need internet access for the initial software downloads and GitHub account creation. Once this PDF and the installers are available, the Java compile/run checks themselves can be completed locally.

1. Mandatory: What You Must Bring / Have Ready

Requirement	Ready when...
☐ Laptop + charger	Bring your own laptop. Make sure you can install software on it (administrator access may be required).
☐ Working Java JDK 17	Both the java runtime and javac compiler must work from Command Prompt/Terminal.
☐ Git installed	The git command must work. Git will be used to work with GitHub repositories.
☐ GitHub account	Create a personal GitHub account, verify the email address, and make sure you can sign in.
☐ Know your login details	Do not arrive unable to access your GitHub account or university email.

Course standard: Use JDK 17 for this course so that everyone compiles and runs code with a consistent Java version. An IDE is not required for this setup check; the first test is intentionally done from Command Prompt/Terminal.

2. Install Java JDK 17

1

Download a JDK 17 installer

Use a trusted JDK distribution. For Windows, Eclipse Temurin 17 (JDK, x64, MSI) is a straightforward option.

Official Temurin downloads: <https://adoptium.net/temurin/releases/?version=17>

2

Install the JDK

Run the installer. Keep the normal/default components. If the installer offers options to add Java to PATH or set JAVA_HOME, enable them.

3 Close and reopen Command Prompt

Environment-variable changes may not be visible in a terminal window that was already open before installation.

4 Verify both Java commands

Open Command Prompt (Windows) or Terminal (macOS/Linux) and run:

```
java --version
javac --version
```

PASS condition: both commands display version information and indicate Java 17. If java works but javac does not, your compiler/JDK setup is incomplete.

3. Test Java Without an IDE

This test confirms the complete path: source file -> Java compiler -> bytecode (.class) -> JVM execution.

1 Create a simple working folder

For example, create a folder named CSC241-Setup on your Desktop and open Command Prompt/Terminal in that folder.

2 Create HelloWorld.java

Create a plain-text file named exactly HelloWorld.java and enter the following code:

```
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Java setup is working!");
    }
}
```

Filename rule: Because the class is public and named HelloWorld, the source file must be named HelloWorld.java.

3 Compile the source file

Run:

```
javac HelloWorld.java
```

Expected result: if there is no compilation error, a file named HelloWorld.class is created in the same folder.

4 Run the compiled program

Run the class name without .java or .class:

```
Expected output: Java setup is working!  
java HelloWorld
```

Do not stop after java --version. A successful HelloWorld compile-and-run test is the best proof that the JDK, compiler and execution environment are ready.

4. Install Git

Git is the version-control tool that will let you work with repositories hosted on GitHub.

1

Install Git for your operating system

Windows students can use the official Git for Windows installer. Default installer choices are suitable for this course unless your instructor says otherwise.

Official Git download: <https://git-scm.com/downloads>

2

Verify Git

Close and reopen Command Prompt/Terminal, then run:

```
git --version
```

PASS condition: Git prints a version number instead of "command not found" or "not recognized".

5. Create and Verify Your GitHub Account

1

Create a personal GitHub account

If you already have one, do not create another account. Use a professional username that you are comfortable using for coursework.

GitHub sign-up: <https://github.com/signup>

2

Verify your email address

GitHub requires a verified email address for some basic tasks such as creating repositories. Complete the verification message sent to your email.

3

Test sign-in before the lab

Sign out once and sign in again. You should know your username and be able to access the account without resetting your password during the lab.

Security: Use a strong unique password. Enabling two-factor authentication (2FA) is strongly recommended. Never share your GitHub password or verification codes with anyone.

6. Quick Troubleshooting

Problem	Likely cause	What to do
java is not recognized	Java not installed or PATH not updated	Install JDK 17; close/reopen terminal. On Windows, use where java to see what executable is being found.
javac is not recognized	Compiler/JDK not on PATH	Confirm you installed a JDK, not only a runtime. Re-run installer or correct PATH.
Wrong Java version appears	Another Java installation is earlier on PATH	On Windows run where java and where javac. Remove/correct conflicting PATH entries or ask during lab.
Could not find or load main class	Wrong folder/class name/command	Make sure Command Prompt is in the folder containing HelloWorld.class and run java HelloWorld.
Compilation error in HelloWorld	Typing/file-name error	Compare the code carefully. Ensure file is HelloWorld.java and quotation marks/braces are correct.
git is not recognized	Git not installed or terminal not refreshed	Install Git and open a new terminal window, then run git --version again.
Cannot create GitHub repository	Email/account verification incomplete	Open GitHub account settings/email and complete verification.

7. Final Readiness Check - Do This Before Leaving Home

- ☐ I have my laptop and charger.
- ☐ I can run java --version and it shows Java 17.
- ☐ I can run javac --version and it shows Java 17.
- ☐ I successfully compiled HelloWorld.java and ran java HelloWorld.
- ☐ I can run git --version.
- ☐ I have a GitHub account with a verified email address.
- ☐ I can sign in to GitHub and I know my username.
- ☐ I have saved this setup guide on my laptop/phone so I can use it even if internet access is unavailable during setup/troubleshooting.

READY FOR LAB means: Java compiles and runs from the command line, Git works, and you can sign in to a verified GitHub account. Repository creation and the course-specific Git/GitHub workflow will be practiced in the lab.

Platform note

The screenshots/menu wording of installers may differ across Windows, macOS and Linux. The verification commands **java --version**, **javac --version**, **git --version** and the HelloWorld compile/run steps are the same in principle. If you use macOS/Linux, install JDK 17 and Git using the official installer/package method for your operating system.